



**C23-EC-ECII-404**

**23433**

**BOARD DIPLOMA EXAMINATION, (C-23)**  
**MARCH/APRIL—2025**  
**DECE – FOURTH SEMESTER EXAMINATION**  
**IoT AND SENSORS**

*Time : 3 Hours ]*

*[ Total Marks : 80*

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**PART—A**

3×10=30

**Instructions :** (1) Answer **all** questions.  
(2) Each question carries **three** marks.  
(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. List any six goals of industry 4.0.
2. Define IoT.
3. What is meant by actuator in IoT?
4. List any six actuators used in IoT.
5. List any six analog sensors used in IoT.
6. List the specifications of BMP280 pressure sensor module.
7. List the specifications of SigFox technology.
8. List any six applications of GPS.
9. List any three sensors used in IoT based system for smart lighting.
10. Write the uses of IoT based smart irrigation system.

**PART—B**

10×5=50

**Instructions :** (1) Answer *any five* questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

- 11.** Explain the components of futuristic industrial plant in industry 4.0 with block diagram. 10
- 12.** Explain M2M and IoT technology fundamentals such as : 5+5
- (a) Devices and gateways
- (b) Everything as a service (XaaS)
- 13.** Explain the function of software components in data storage and data analytics in IoT. 10
- 14.** Explain the working principle and state the functions of the following sensors.
- (a) Proximity sensors 5
- (b) Water quality sensors 5
- 15.** Explain different mobile technologies (2G, 3G, 5G) used in IoT. 10
- 16.** What are the differences between Bluetooth and Wi-Fi technologies? 10
- 17.** Explain the IoT based system for home automation with a block diagram. 10
- 18.** Explain the IoT based system for healthcare with a block diagram. 10

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